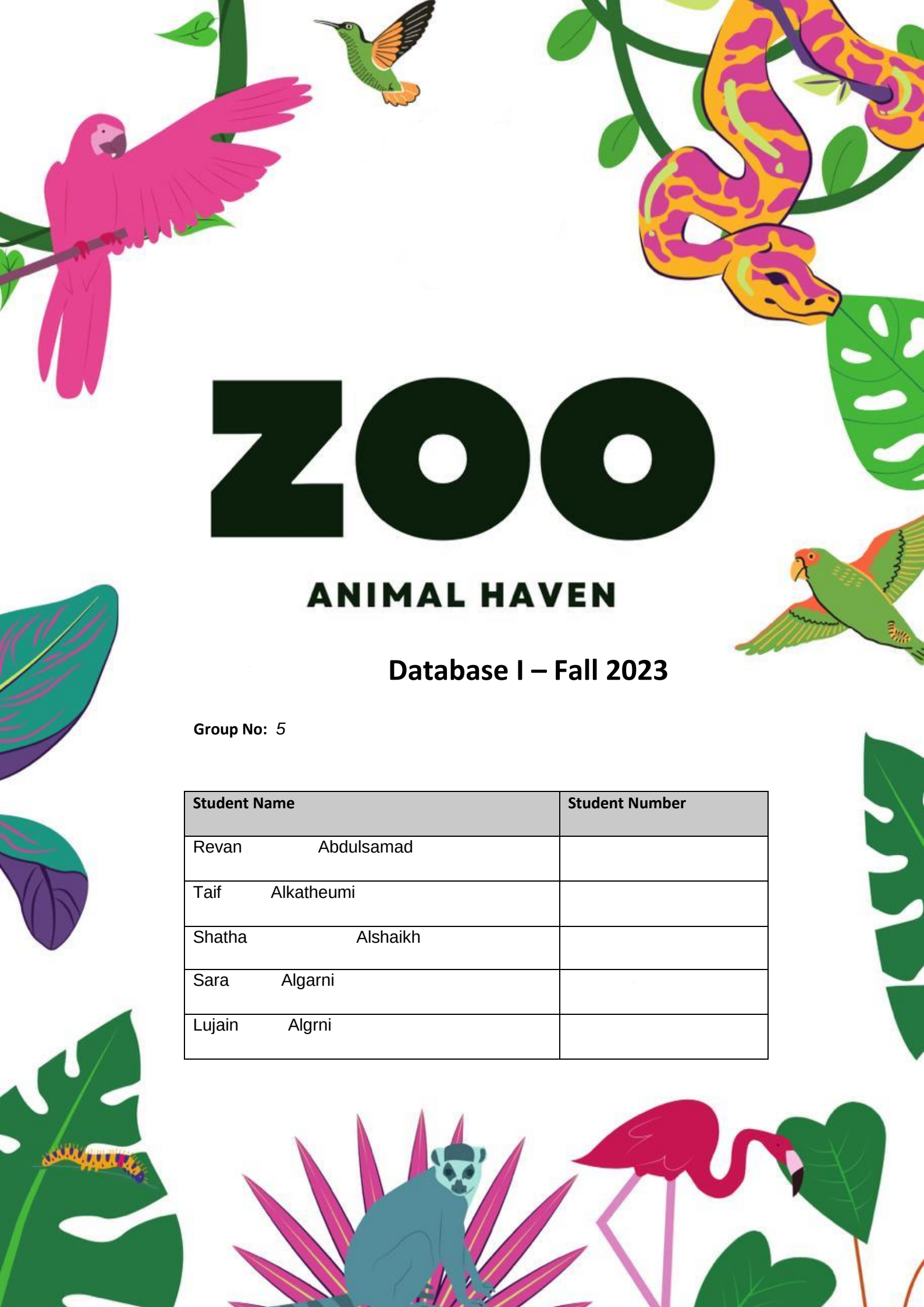




ZOO

ANIMAL HAVEN

Database I – Fall 2023

Group No: 5

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PART I: Analysis

1 Problem Definition and Data Requirements

1.1 Problem Definition

The world of animals is a world full of wonder, beauty, and strangeness. The animal kingdom is one of the most diverse biological kingdoms and includes different land types. These animals live in different parts of the world and are distributed in different regions depending on the structure and properties of each animal. Therefore, it is difficult for humans to see multiple animals gathered in one place.

The zoo is a great investment opportunity, and to run it successfully, it must have an ideal database management system that stores and processes data. Furthermore, it must pay attention to visitor's requirements to ensure their satisfaction with the services provided by the zoo.

Our goal is to develop and implement a comprehensive and easy-to-use database system. This system must be able to efficiently store, manage, and analyze different types of data related to animals, areas, departments, employees, visitors, bills, ticket and events. By having a well-designed database system, a zoo can streamline its operations, enhance data accessibility and accuracy, improve decision-making processes, improve resource management, and provide a better visitor experience.

1.2 Data Requirements

Employee:

- Emp_ID: unique number
- Name (EmpFName, EmpLName)
- Position
- E_Phone
- E_Gender
- E_DateOfBirth
- Age
- Salary

Visitor:

- VisitorPhone: unique number
- VisitorName (VisitorFName, VisitorLName)

Department:

- DepartmentID: unique number
- DepartmentName (Guidance, Areas services, Event management,...etc.)
- DepartmentLocation

Area:

- AreaID: unique number
- AreaLocation
- AreaName (Aquarium, Farm animals and pets, Predators,...etc.)

Animal:

- AnimalID: unique number
- Type
- A_Gender
- A_DateOfBirth

Event:

- EventID: unique number
- EventLocation
- EventName

Bill:

- Bill_ID: unique number
- TotalPrice
- Status

Payment Method:

- MethodID: unique number
- MethodName (Master card, Apple pay, Cash, PayPal)

Area Ticket:

- AreaTID: unique number
- Date
- Price
- TicketName

Event Ticket

- EventTID: unique number
- Date
- Price
- TicketName

1.2 Business Rules

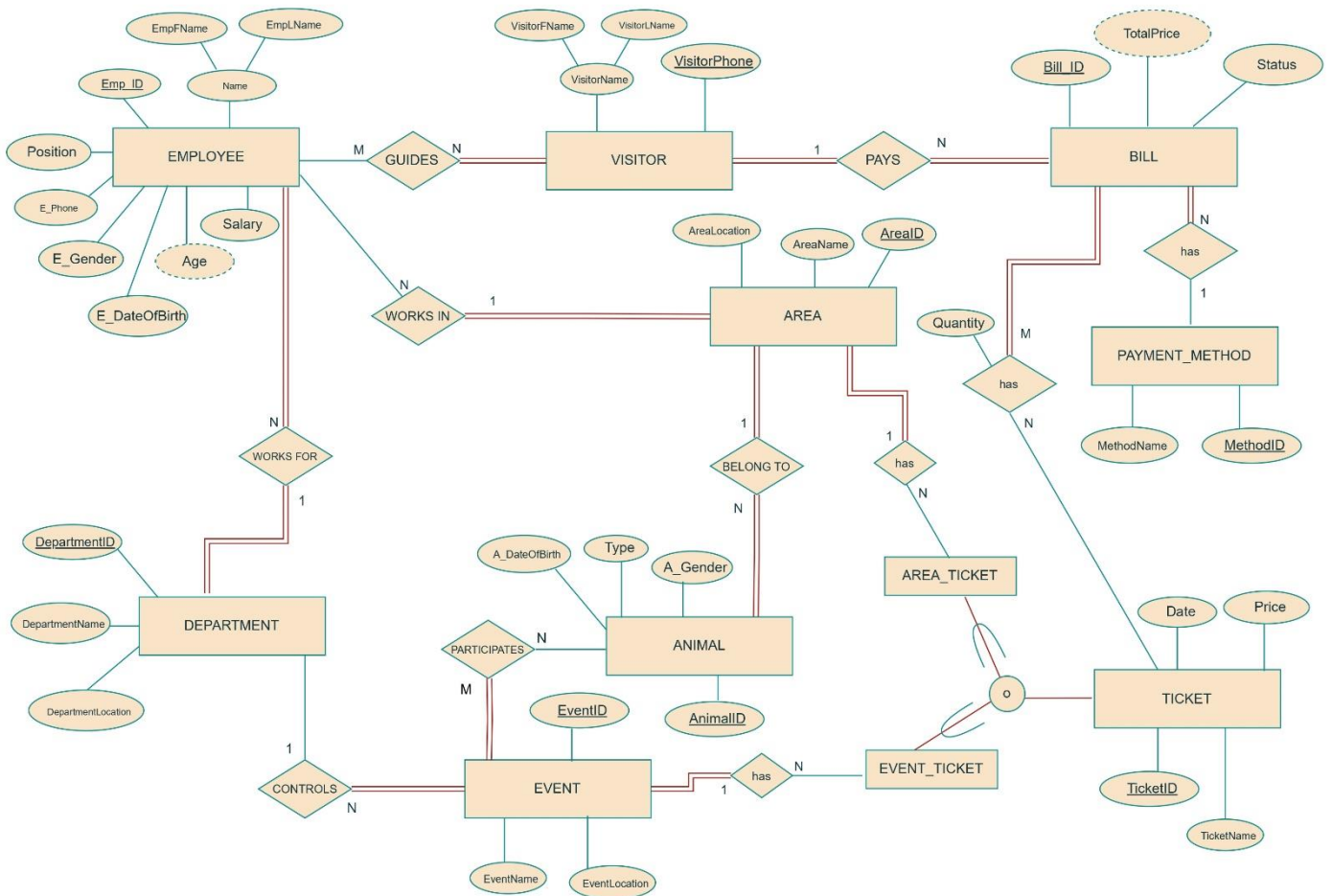
To manage the Animal Haven Zoo efficiently without any conflicts, there are some rules specified and stated by the owner that we must take into consideration while designing the database.

- Each employee must work for one department, and each department must have at least one employee belonging to it.
- An employee guides many visitors, and each visitor can be guided by one or more employees.
- An employee can work in one area, and each area must have many employees.
- Employee's age must be greater than or equal to 18 years.
- An area must have several animals, and each animal must belong to one area.
- Each event must have one location.
- Each visitor must pay one or more bills, and each bill must be paid by one visitor.
- A department controls several events, and each event must be controlled by a department.
- An animal can participate in several events, and each event must have one or more animals participating in it.
- Each bill must have a payment method, and each payment method can have one or more bills.
- Each bill must have many tickets, and each ticket can have one or more bills.
- An area must have several area tickets, and each area ticket can have only one area.
- An event must have several event tickets, and each event ticket can have only one event.

PART II: DB Design

2 ER Diagram Design

2.1 ER diagram



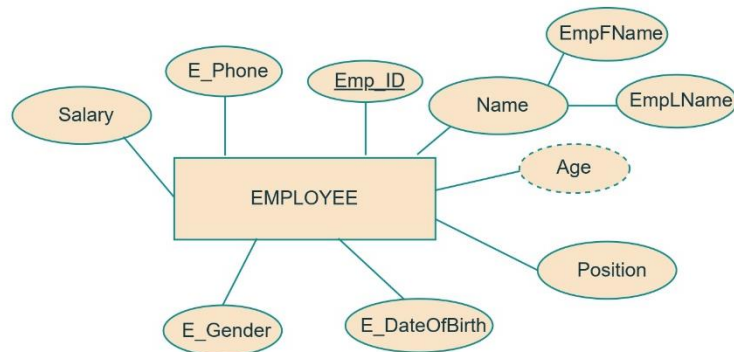
2.2 Design of Business Rules

Business Rule	Design Decisions	Justification
Each employee must work for one department, and each department must have at least one employee belonging to it.	1: N Binary relationship between department and employee.	Both the employee and department entities have total participation in the relationship "Works for".
An employee guides many visitors, and each visitor can be guided by one or more employees.	N: M binary relationship between employee and visitor.	The employee entity has partial participation with the relationship "Guides " while the visitor entity has total participation.
An employee can work in one area, and each area must have many employees.	1: N Binary relationship between area and employee.	The employee entity has partial participation with the relationship "Works in " while the area entity has total participation.
Employee's age must be greater than or equal to 18 years.	A derived attribute called Age will be calculated from the date of birth and the current date.	If the calculated age value is under 18 then this employee instance validates the age constraint which will be handled in the implementation phase.
An area must have several animals, and each animal must belong to one area.	1: N binary relationship between area and animal.	Both area and animal entities have total participation with the relationship "Belongs to".
Each event must have one location.	This rule represents (not null) constraints in the implementation phase.	-
A department controls several events, and each event must be controlled by a department.	1: N binary relationships between department and event.	The department entity has partial participation with the relationship "controls" while the event entity has total participation.
Each visitor must pay one or more bills, and each bill must be paid by one visitor .	1: N binary relationships between visitor and bill.	Both visitor and bill entities have total participation with the relationship "Pays".

An animal can participate in several events, and each event must have one or more animals participating in it.	N:M binary relationship between animal and event.	The animal entity has partial participation with the relationship "Participates" while the event entity has total participation.
Each bill must have a payment method, and each payment method can have one or more bills.	1:N binary relationship between payment method and bill.	Payment method entity has partial participation with the relationship "has" while the bill entity has total participation.
Each bill must have many tickets, and each ticket can have one or more bills.	N:M binary relationship between bill and ticket.	The ticket entity has partial participation with the relationship "has" while the bill entity has total participation.
An area must have several area tickets, and each area ticket can have only one area.	1: N binary relationships between area and area_ticket.	The area_ticket entity has partial participation with the relationship "has" while the area entity has total participation.
An event must have several event tickets, and each event ticket can have only one event.	1: N binary relationships between event and event_ticket.	The event_ticket entity has partial participation with the relationship "has" while the area entity has total participation.

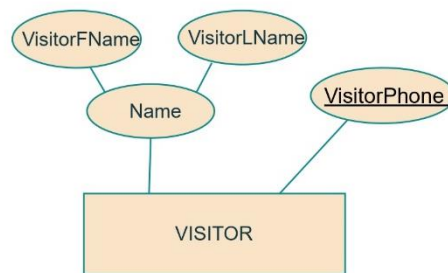
3 ER-to-Logical Schema Mapping

3.1 Mapping of Regular Entity Types



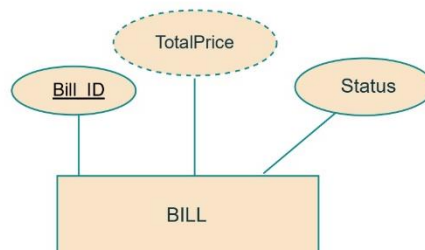
EMPLOYEE

<u>Emp_ID</u>	EmpFName	EmpLName	E_DateOfBirth	Age	Position	Salary	E_Gender	E_Phone
---------------	----------	----------	---------------	-----	----------	--------	----------	---------



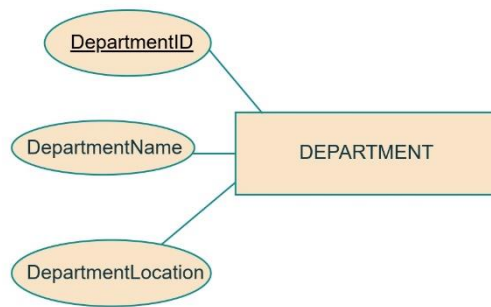
VISITOR

<u>VisitorPhone</u>	VisitorFName	VisitorLName
---------------------	--------------	--------------



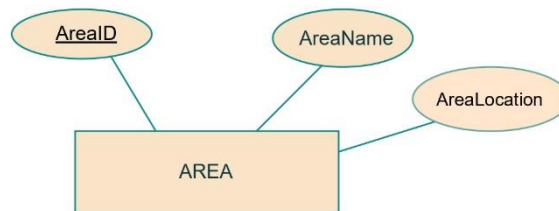
BILL

<u>Bill_ID</u>	Status	TotalPrice
----------------	--------	------------



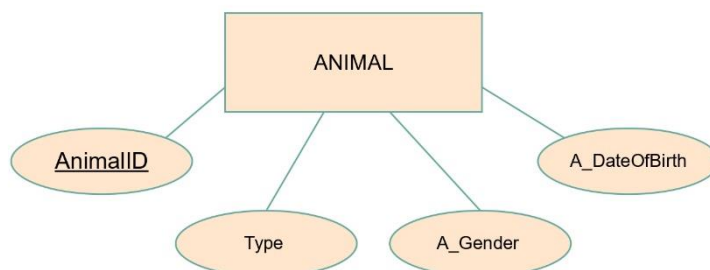
DEPARTMENT

<u>DepartmentID</u>	DepartmentName	DepartmentLocation
---------------------	----------------	--------------------



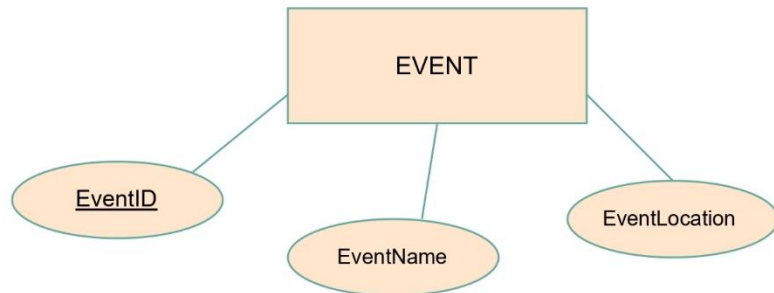
AREA

<u>AreaID</u>	AreaName	AreaLocation
---------------	----------	--------------



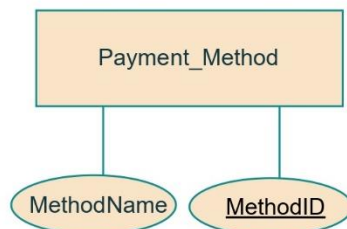
ANIMAL

<u>AnimalID</u>	Type	A_Gender	A_DateOfBirth
-----------------	------	----------	---------------



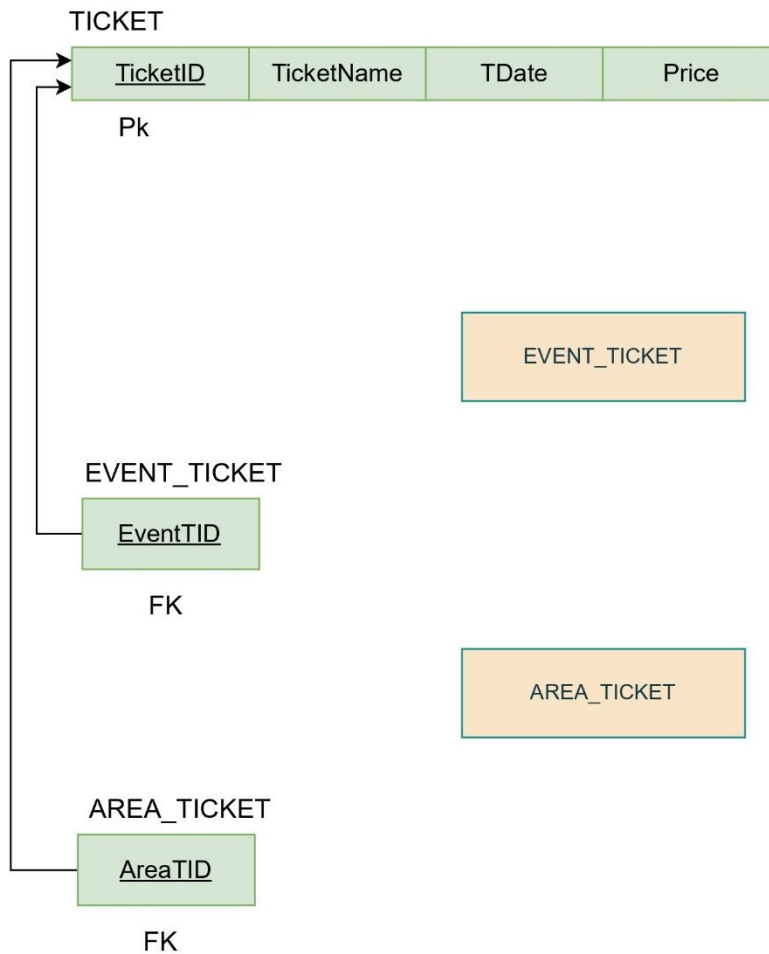
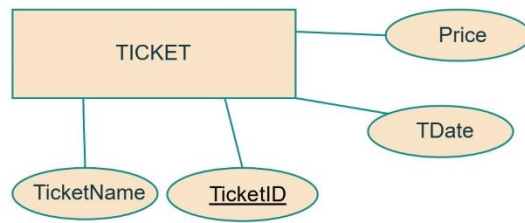
EVENT

<u>EventID</u>	EventName	EventLocation
----------------	-----------	---------------



Paymen_tMethod

<u>MethodID</u>	MethodName
-----------------	------------



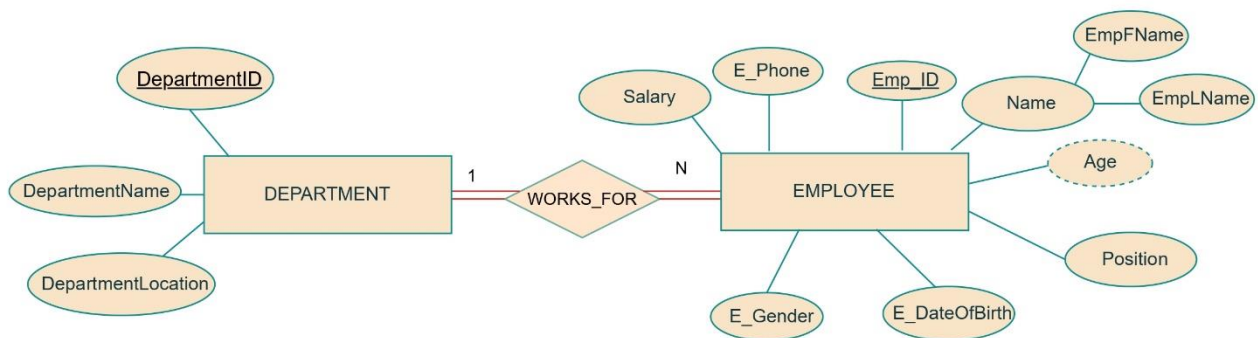
3.2 Mapping of Weak Entity Types

The system doesn't have any weak entity.

3.3 Mapping of Binary 1-1 Relationship Types

The system doesn't have any binary 1-1 relationship types.

3.4 Mapping of Binary 1-N Relationship Types



DEPARTMENT

<u>DepartmentID</u>	DepartmentName	DepartmentLocation
---------------------	----------------	--------------------

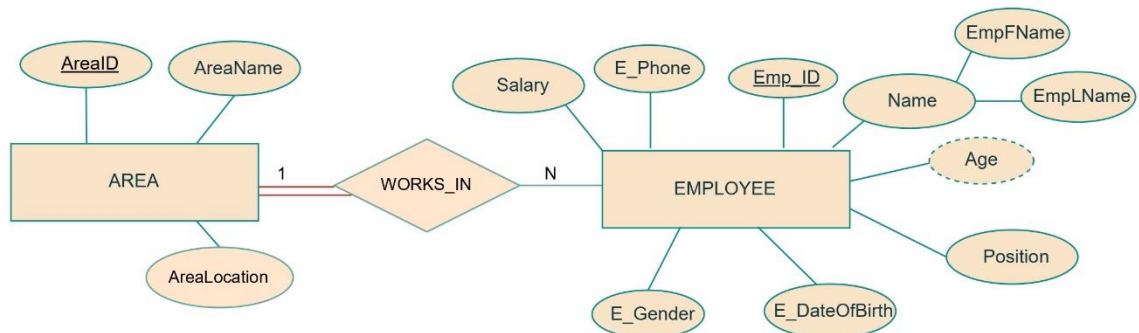
Pk

EMPLOYEE

<u>Emp_ID</u>	EmpFName	EmpLName	E_DateOfBirth	Age	Position	Salary	E_Gender	E_Phone	Dep_ID
---------------	----------	----------	---------------	-----	----------	--------	----------	---------	--------

Pk

Fk



AREA

<u>AreaID</u>	AreaName	AreaLocation
---------------	----------	--------------

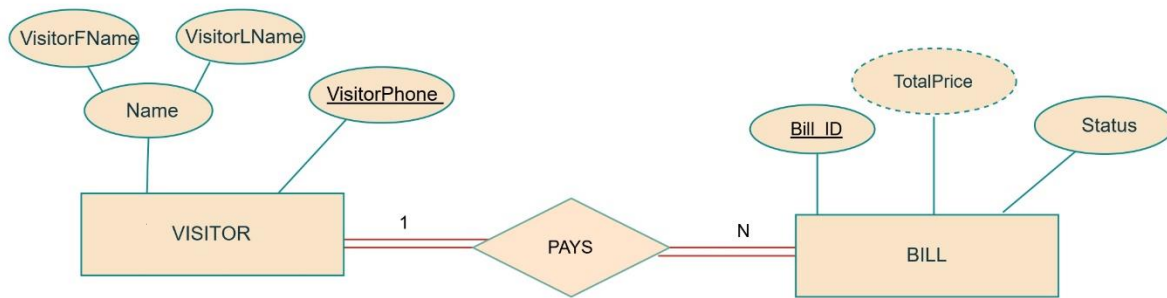
Pk

EMPLOYEE

<u>Emp_ID</u>	EmpFName	EmpLName	E_DateOfBirth	Age	Position	Salary	E_Gender	E_Phone	Dep_ID	Area_ID
---------------	----------	----------	---------------	-----	----------	--------	----------	---------	--------	---------

Pk

Fk



VISITOR

<u>VisitorPhone</u>	VisitorFName	VisitorLName
---------------------	--------------	--------------

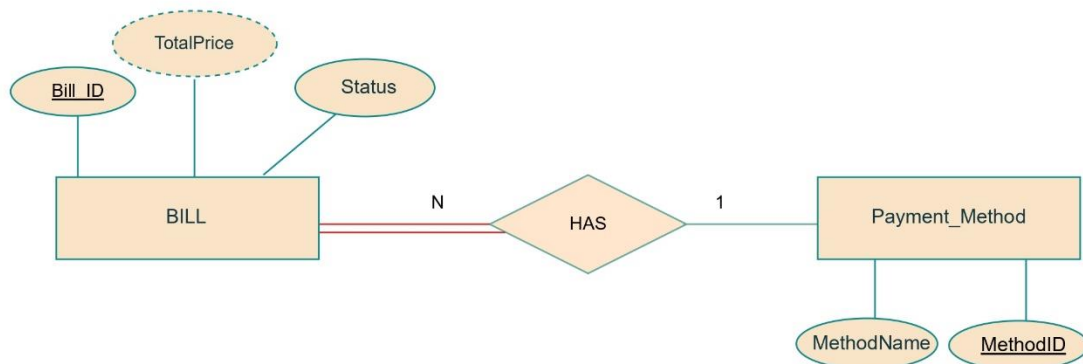
Pk

BILL

<u>Bill_ID</u>	Status	TotalPrice	Visitor_Phone
----------------	--------	------------	---------------

Pk

Fk



PAYMENT_METHOD

<u>MethodID</u>	MethodName
-----------------	------------

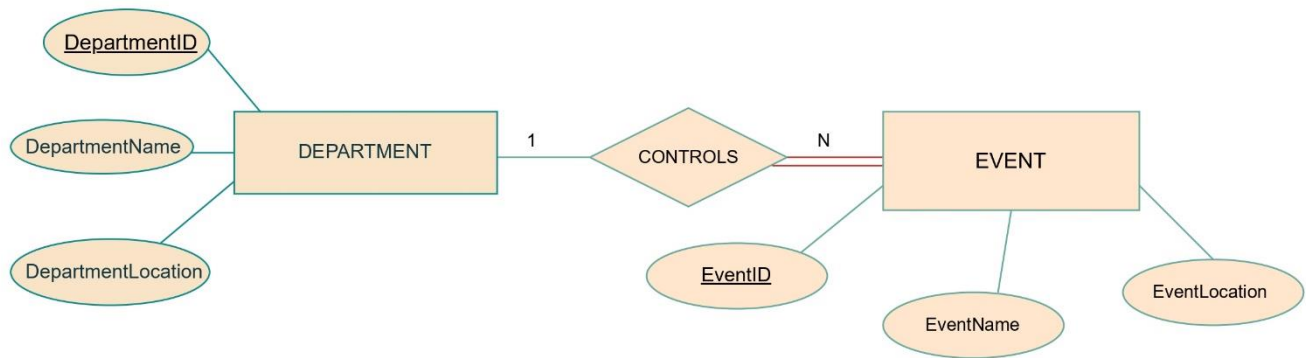
Pk

BILL

<u>Bill_ID</u>	Status	TotalPrice	Visitor_Phone	PMethod_ID
----------------	--------	------------	---------------	------------

Pk

Fk



DEPARTMENT

<u>DepartmentID</u>	DepartmentName	DepartmentLocation
---------------------	----------------	--------------------

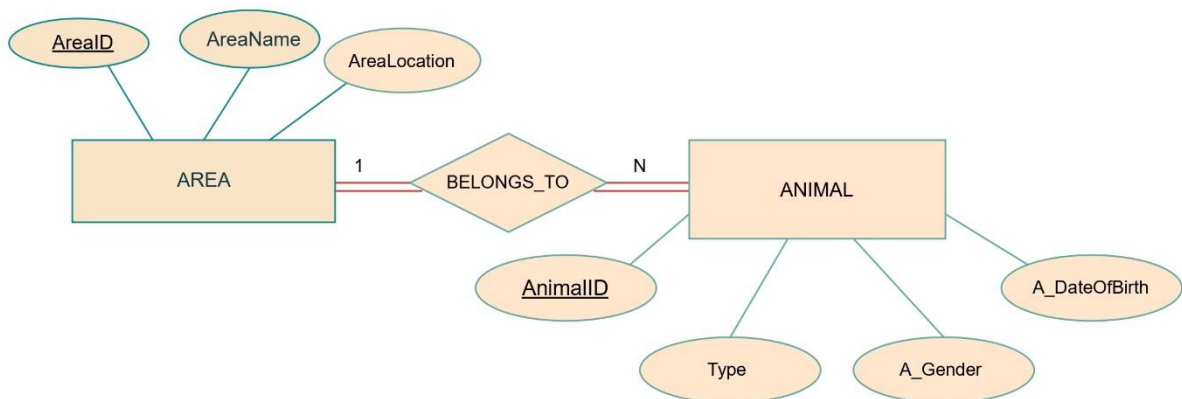
Pk

EVENT

<u>EventID</u>	EventName	EventLocation	Dep_ID
----------------	-----------	---------------	--------

Pk

Fk



AREA

<u>AreaID</u>	AreaName	AreaLocation
---------------	----------	--------------

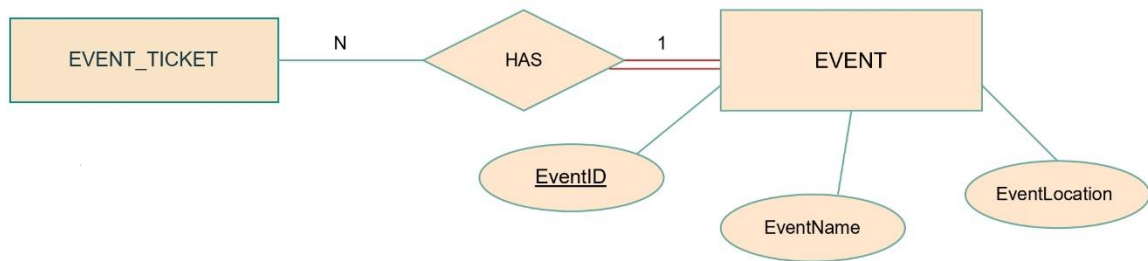
Pk

ANIMAL

<u>AnimalID</u>	Type	A_Gender	A_DateOfBirth	Area_ID
-----------------	------	----------	---------------	---------

Pk

Fk



EVENT

<u>EventID</u>	EventName	EventLocation	Dep_ID
----------------	-----------	---------------	--------

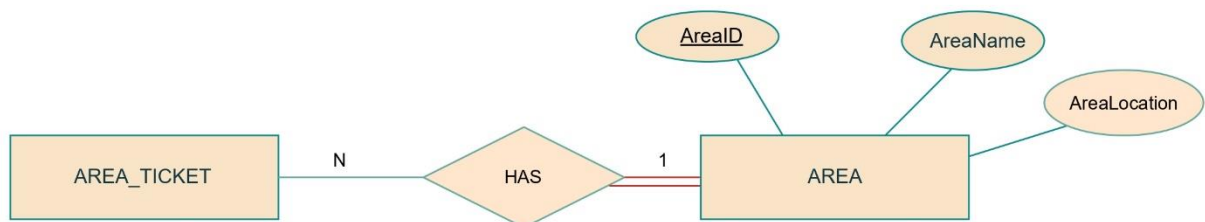
Pk

EVENT_TICKET

<u>EventTID</u>	Event_ID
-----------------	----------

Pk

Fk



AREA

<u>AreaID</u>	AreaName	AreaLocation
---------------	----------	--------------

Pk

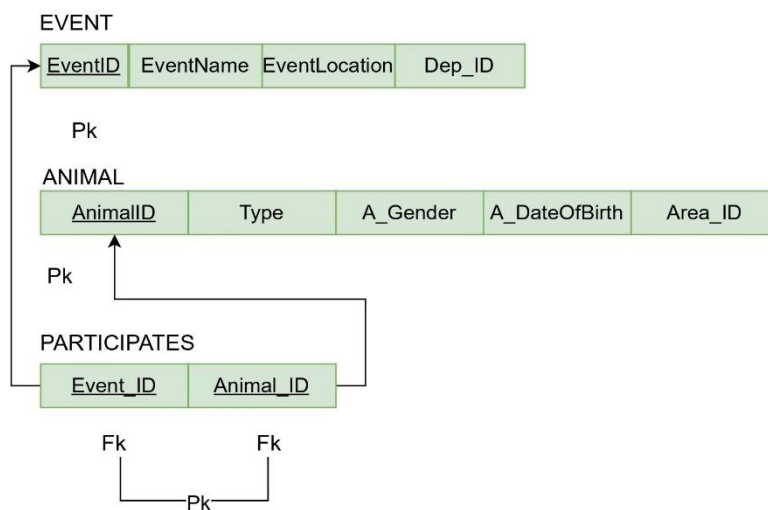
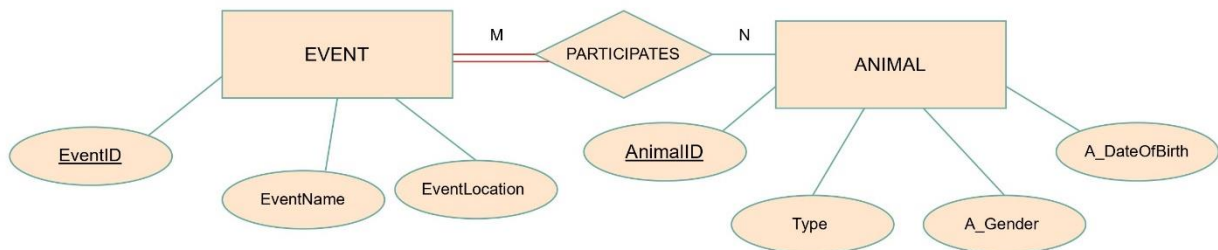
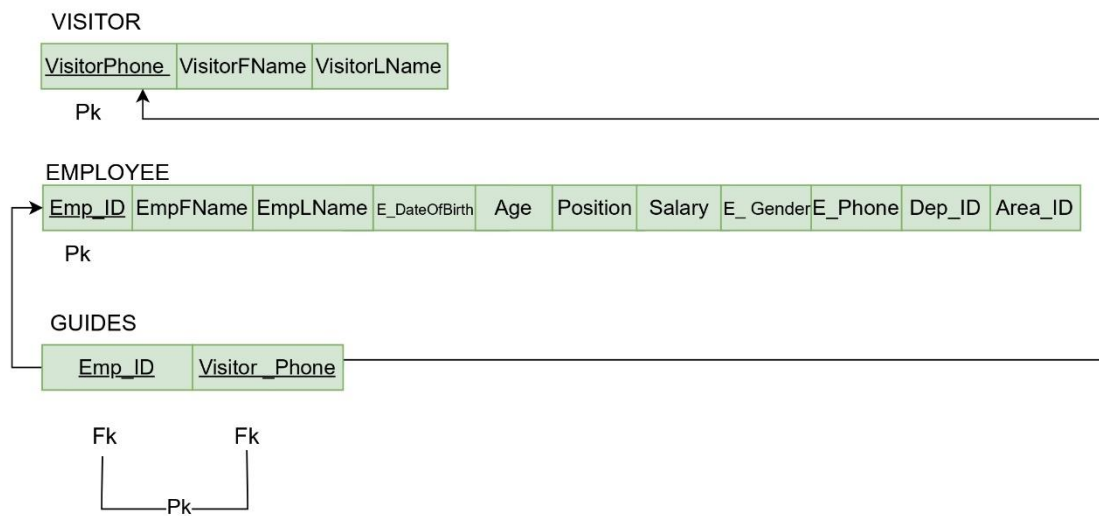
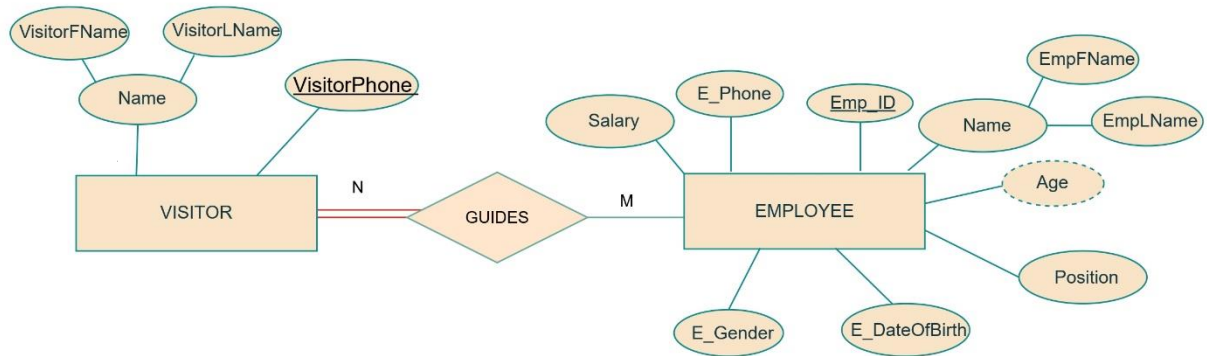
AREA_TICKET

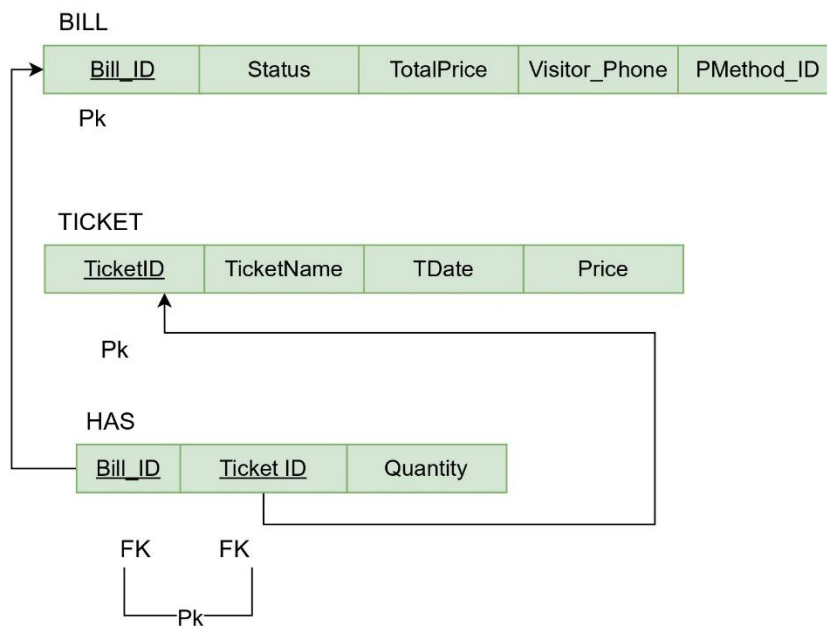
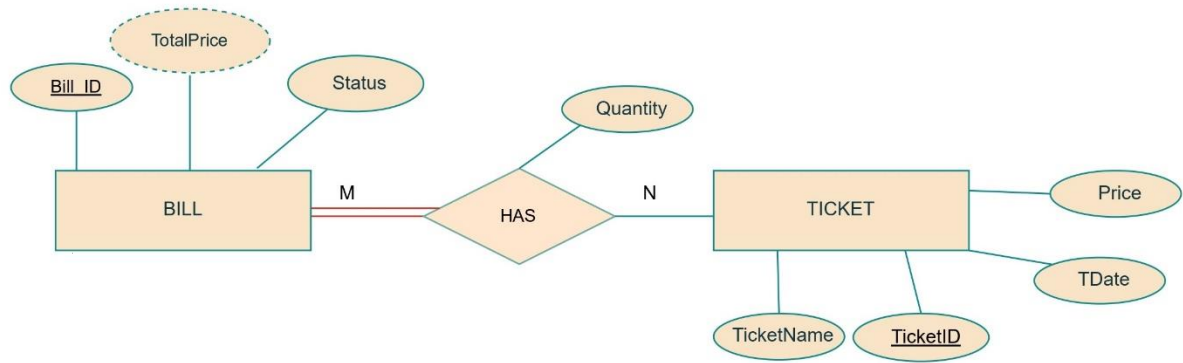
<u>AreaTID</u>	Area_ID
----------------	---------

Pk

Fk

3.5 Mapping of Binary M-N Relationship Types





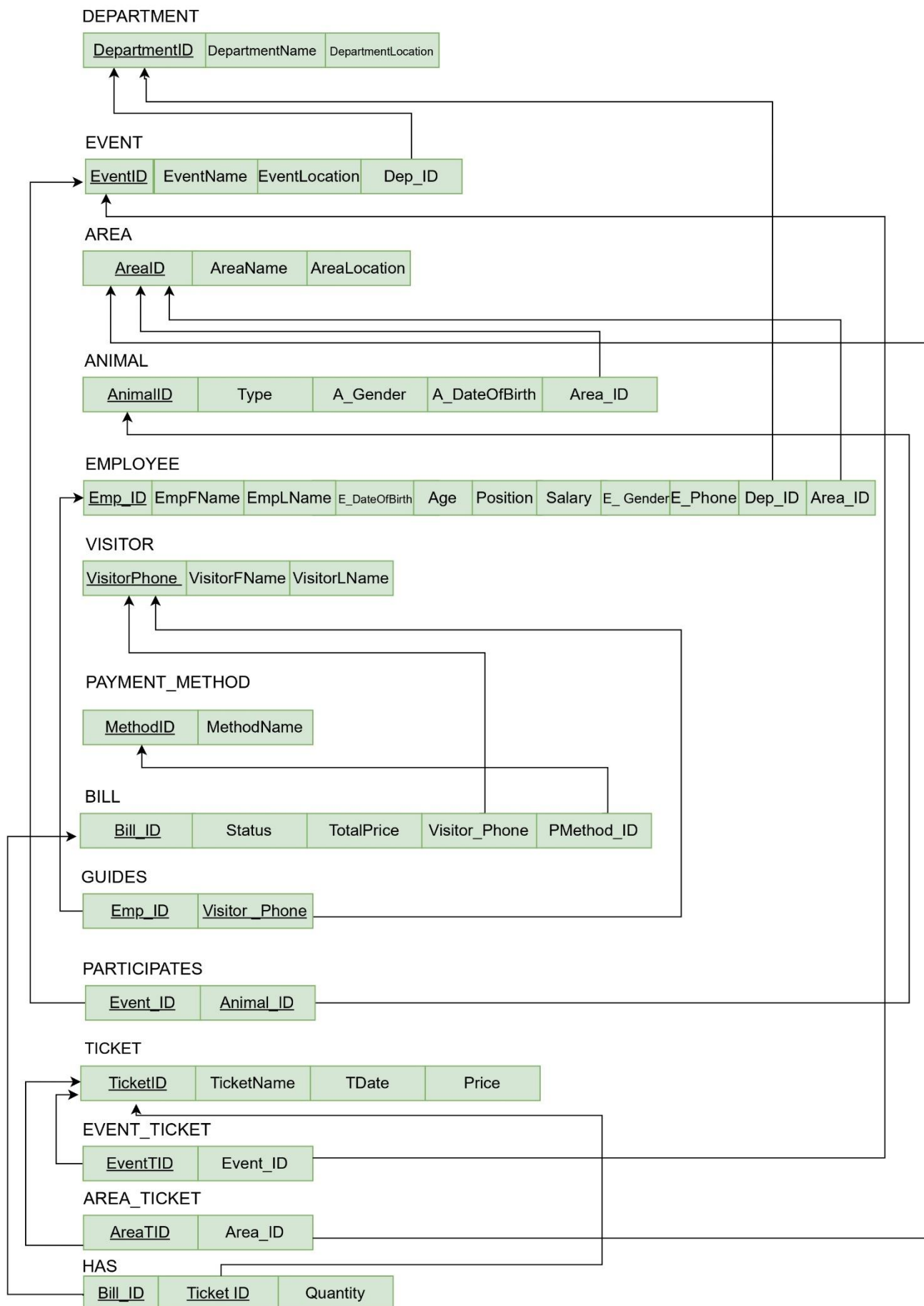
3.6 Mapping of Multivalued Attributes

The system doesn't have any multivalued attributes.

3.7 Mapping of n-ary Relationship Types

The system doesn't have any n-ary relationship.

3.8 Schema Diagram



4 Normalization

4.1 First Normal Form

The following schemas are in the first normal form because they do not contain multivalued, composite attributes or nested relations.

DEPARTMENT

<u>DepartmentID</u>	DepartmentName	DepartmentLocation
---------------------	----------------	--------------------

EVENT

<u>EventID</u>	EventName	EventLocation	Dep_ID
----------------	-----------	---------------	--------

AREA

<u>AreaID</u>	AreaName	AreaLocation
---------------	----------	--------------

ANIMAL

<u>AnimalID</u>	Type	A_Gender	A_DateOfBirth	Area_ID
-----------------	------	----------	---------------	---------

EMPLOYEE

<u>Emp_ID</u>	EmpFName	EmpLName	E_DateOfBirth	Age	Position	Salary	E_Gender	E_Phone	Dep_ID	Area_ID
---------------	----------	----------	---------------	-----	----------	--------	----------	---------	--------	---------

VISITOR

<u>VisitorPhone</u>	VisitorFName	VisitorLName
---------------------	--------------	--------------

PAYMENT_METHOD

<u>MethodID</u>	MethodName
-----------------	------------

BILL

<u>Bill_ID</u>	Status	TotalPrice	Visitor_phone	PMethod_ID
----------------	--------	------------	---------------	------------

GUIDES

<u>Emp_ID</u>	<u>Visitor_Phone</u>
---------------	----------------------

PARTICIPATES

<u>Event_ID</u>	<u>Animal_ID</u>
-----------------	------------------

TICKET

<u>TicketID</u>	TicketName	TDate	Price
-----------------	------------	-------	-------

EVENT_TICKET

<u>EventTID</u>	Event_ID
-----------------	----------

AREA_TICKET

<u>AreaTID</u>	Area_ID
----------------	---------

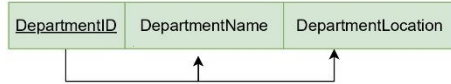
HAS

<u>Bill_ID</u>	<u>Ticket_ID</u>	Quantity
----------------	------------------	----------

4.2 Second Normal Form

The following schemas are in the second normal form because all non-prime attributes are fully dependent on the primary key, so there are no partial dependencies.

DEPARTMENT



EVENT



AREA



ANIMAL



EMPLOYEE



VISITOR



PAYMENT_METHOD



BILL

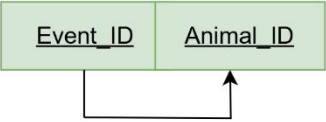


The following schemas are in the second normal form because there are no non prime attributes in the relation. Therefore, it will always be in 2NF.

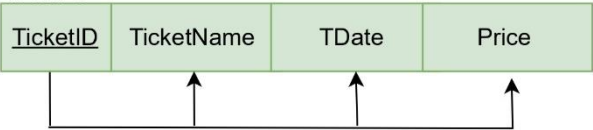
GUIDES



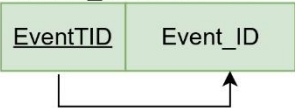
PARTICIPATES



TICKET



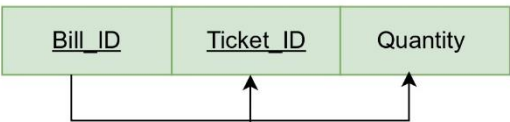
EVENT_TICKET



AREA_TICKET



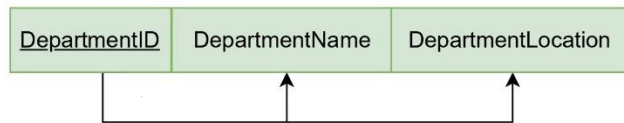
HAS



4.3 Third Normal Form

1-The following schemas are in the third normal form because there are no transitive dependencies, all non-prime attributes depend on nothing but the key.

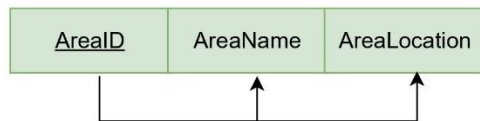
DEPARTMENT



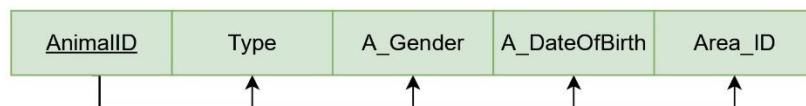
EVENT



AREA



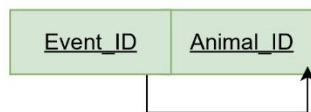
ANIMAL



VISITOR



PARTICIPATES



PAYMENT_METHOD



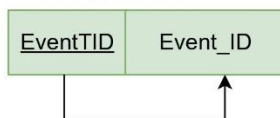
BILL



TICKET



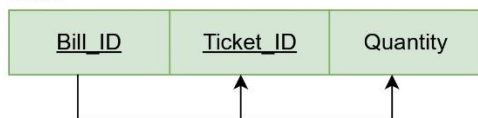
EVENT_TICKET



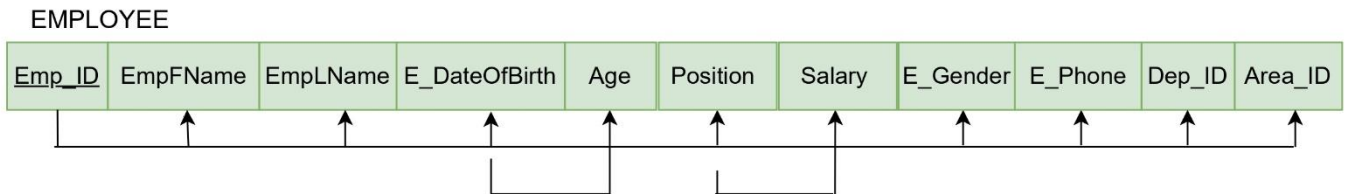
AREA_TICKET



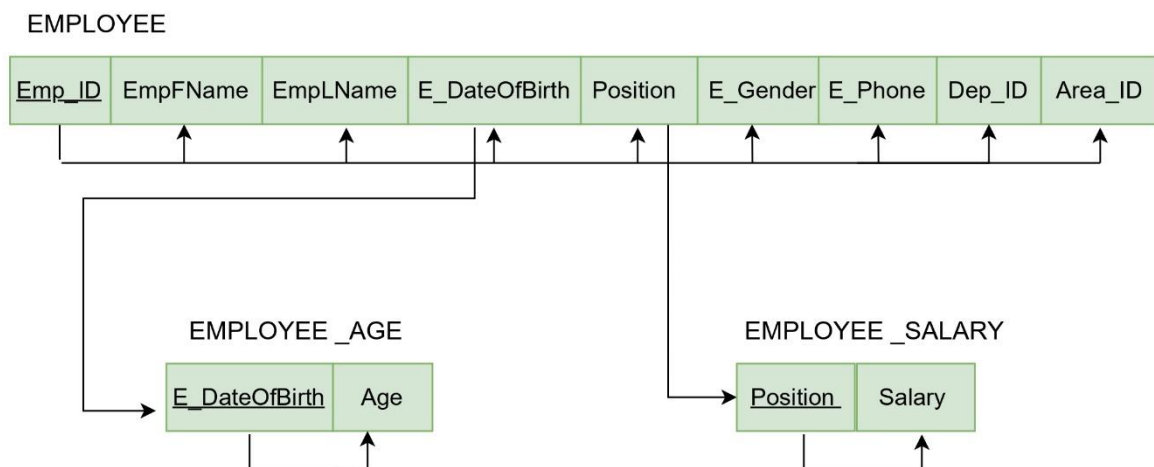
HAS



2-The following schemas are not in the third normal form since not all non prime attributes depend on the key, and there are transitive dependencies.



To achieve 3NF, we need to eliminate transitive dependencies by creating additional tables

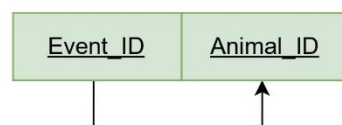


3-The following schemas are in the third normal form because there are no non prime attributes ensuring that there are no transitive dependencies.

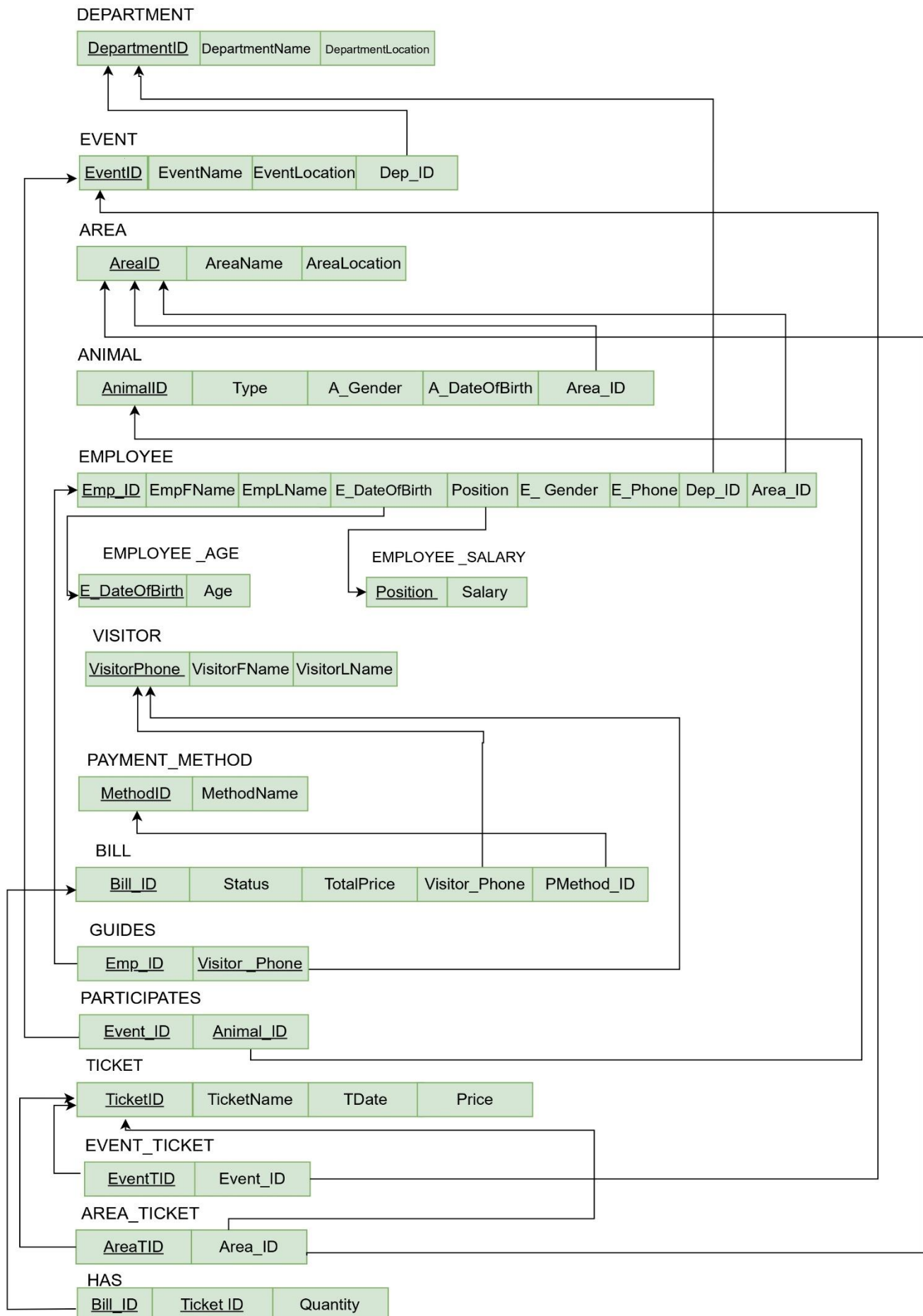
GUIDES



PARTICIPATES



5 Final DB Schema Diagram



PART III: Implementation

6 Table Creation Script

6.1 <DEPARTMENT> Table

```
CREATE TABLE DEPARTMENT(DepartmentID NUMBER(3) PRIMARY KEY,  
                           DepartmentName VARCHAR2(50),  
                           DepartmentLocation VARCHAR2(50));
```

```
SQL> CREATE TABLE DEPARTMENT(DepartmentID NUMBER(3) PRIMARY KEY,  
 2      DepartmentName VARCHAR2(50),  
 3      DepartmentLocation VARCHAR2(50));  
  
Table created.
```

6.2 <EVENT> Table

```
CREATE TABLE EVENT(EventID NUMBER(3) PRIMARY KEY,  
                    EventName VARCHAR2(50),  
                    EventLocation VARCHAR2(50) NOT NULL,  
                    Dep_ID NUMBER(3),
```

```
CONSTRAINT FK_EVNDP
```

```
FOREIGN KEY (DEP_ID) REFERENCES DEPARTMENT(DepartmentID) ON  
DELETE CASCADE);
```

```
SQL> CREATE TABLE EVENT(EventID NUMBER(3) PRIMARY KEY,  
 2      EventName VARCHAR2(50),  
 3      EventLocation VARCHAR2(50) not null ,  
 4      Dep_ID NUMBER(3),  
 5      CONSTRAINT FK_EVNDP  
 6      FOREIGN KEY (DEP_ID) REFERENCES DEPARTMENT(DepartmentID) ON DELETE CASCADE);  
  
Table created.
```

6.3 <AREA> Table

```
CREATE TABLE AREA(Area_ID NUMBER(3) PRIMARY KEY,  
                   AreaName VARCHAR2(50),  
                   AreaLocation VARCHAR2(50));
```

```
SQL> CREATE TABLE AREA(Area_ID NUMBER(3) PRIMARY KEY,  
2                          AreaName VARCHAR2(50),  
3                          AreaLocation VARCHAR2(50));  
  
Table created.
```

6.4 <ANIMAL> Table

```
CREATE TABLE ANIMAL(AnimalID NUMBER(3) PRIMARY KEY,  
                     Type VARCHAR2(10),  
                     A_Gender VARCHAR2(10),  
                     A_DateOfBirth DATE,  
                     Area_ID NUMBER(3),  
  
CONSTRAINT FK_ANMARE  
FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE);
```

```
SQL> CREATE TABLE ANIMAL(AnimalID NUMBER(3) PRIMARY KEY,  
2                          Type VARCHAR2(10),  
3                          A_Gender VARCHAR2(10),  
4                          A_DateOfBirth DATE,  
5                          Area_ID NUMBER(3),  
6  CONSTRAINT FK_ANMARE  
7  FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE);  
  
Table created.
```

6.5 <EMPLOYEE_AGE> Table

```
CREATE TABLE EMPLOYEE_AGE(E_DateOfBirth DATE,  
    Age NUMBER(2) CHECK (Age>18),  
CONSTRAINT PK_EMPLOYEE_AGE PRIMARY KEY (E_DateOfBirth));
```

```
SQL> CREATE TABLE EMPLOYEE_AGE(E_DateOfBirth DATE,  
    2      Age NUMBER(2) CHECK (Age>18),  
    3  CONSTRAINT PK_EMPLOYEE_AGE PRIMARY KEY (E_DateOfBirth));  
  
Table created.
```

6.6 <EMPLOYEE_SALARY> Table

```
CREATE TABLE EMPLOYEE_SALARY(Position VARCHAR2(30),  
    Salary NUMBER(5),  
CONSTRAINT PK_EMPLOYEE_SALARY PRIMARY KEY (Position));
```

```
SQL> CREATE TABLE EMPLOYEE_SALARY(Position VARCHAR2(30),  
    2      Salary NUMBER(5),  
    3  CONSTRAINT PK_EMPLOYEE_SALARY PRIMARY KEY (Position));  
  
Table created.
```

6.7 <EMPLOYEE> Table

```
CREATE TABLE EMPLOYEE(Emp_ID NUMBER(3) PRIMARY KEY,  
    EmpFName VARCHAR2(10),  
    EmpLName VARCHAR2(10),  
    E_DateOfBirth DATE,  
    Position VARCHAR2(30),  
    E_Gender VARCHAR2(10),  
    E_Phone NUMBER(10),  
    Dep_ID NUMBER(3),  
    Area_ID NUMBER(3),  
  
    CONSTRAINT FK_EMPDEP  
        FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT(DepartmentID) ON DELETE  
        CASCADE,  
  
    CONSTRAINT FK_EMPARE  
        FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE,  
  
    CONSTRAINT FK_EMPAGE  
        FOREIGN KEY (E_DateOfBirth) REFERENCES EMPLOYEE_AGE(E_DateOfBirth) ON  
        DELETE CASCADE,  
  
    CONSTRAINT FK_EMPSAL  
        FOREIGN KEY (Position) REFERENCES EMPLOYEE_SALARY(Position) ON DELETE  
        CASCADE  
    );
```

```

SQL> CREATE TABLE EMPLOYEE(Emp_ID NUMBER(3) PRIMARY KEY,
2                               EmpFName VARCHAR2(10),
3                               EmpLName VARCHAR2(10),
4                               E_DateOfBirth DATE,
5                               Position VARCHAR2(30),
6                               E_Gender VARCHAR2(10),
7                               E_Phone NUMBER(10),
8                               Dep_ID NUMBER(3),
9                               Area_ID NUMBER(3),
10          CONSTRAINT FK_EMPDEP
11          FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT(DepartmentID) ON DELETE C
ASCAD,
12          CONSTRAINT FK_EMPARE
13          FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE,
14          CONSTRAINT FK_EMPAGE
15          FOREIGN KEY (E_DateOfBirth) REFERENCES EMPLOYEE_AGE(E_DateOfBirth) O
N DELETE CASCADE,
16          CONSTRAINT FK_EMPSAL
17          FOREIGN KEY (Position) REFERENCES EMPLOYEE_SALARY(Position) ON DELET
E CASCADE
18 );

Table created.

```

6.8 <VISITOR> Table

```

CREATE TABLE VISITOR(VisitorPhone NUMBER(10) PRIMARY KEY,
    VisitorFName VARCHAR2(10),
    VisitorLName VARCHAR2(10));

```

```

SQL> CREATE TABLE VISITOR(VisitorPhone NUMBER(10) PRIMARY KEY,
2                               VisitorFName VARCHAR2(10),
3                               VisitorLName VARCHAR2(10));

Table created.

```

6.9 <PAYMENT_METHOD>Table

```
CREATE TABLE PAYMENT_METHOD(MethodID NUMBER(3) PRIMARY KEY,  
                               MethodName VARCHAR2(20));
```

```
SQL> CREATE TABLE PAYMENT_METHOD(MethodID NUMBER(3) PRIMARY KEY,  
2                                     MethodName VARCHAR2(20));  
  
Table created.
```

6.10 <BILL > Table

```
CREATE TABLE BILL(Bill_ID NUMBER(3) PRIMARY KEY,  
                   Status VARCHAR2(10),  
                   TotalPrice NUMBER(5),  
                   Visitor_Phone NUMBER(10),  
                   PMethod_ID NUMBER(3),
```

```
CONSTRAINT FK_BILLVST
```

```
FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON  
DELETE CASCADE,
```

```
CONSTRAINT FK_BILLPM
```

```
FOREIGN KEY (PMethod_ID) REFERENCES PAYMENT_METHOD(MethodID)  
ON DELETE CASCADE);
```

```
SQL> CREATE TABLE BILL(Bill_ID NUMBER(3) PRIMARY KEY,  
2                         Status VARCHAR2(10),  
3                         TotalPrice NUMBER(5),  
4                         Visitor_Phone NUMBER(10),  
5                         PMethod_ID NUMBER(3),  
6                         CONSTRAINT FK_BILLVST  
7                         FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON DELE  
TE CASCADE,  
8                         CONSTRAINT FK_BILLPM  
9                         FOREIGN KEY (PMethod_ID) REFERENCES PAYMENT_METHOD(MethodID) ON DELE  
TE CASCADE);  
  
Table created.
```

6.11 < GUIDES > Table

```
CREATE TABLE GUIDES(Emp_ID NUMBER(3),  
                    Visitor_Phone NUMBER(10),  
  
CONSTRAINT FK_GUEMP  
    FOREIGN KEY (Emp_ID) REFERENCES EMPLOYEE(Emp_ID) ON DELETE  
CASCADE,  
  
CONSTRAINT FK_GUDVST  
    FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON  
DELETE CASCADE,  
  
CONSTRAINT PK_GUEMPVST  
    PRIMARY KEY (Emp_ID,Visitor_Phone));
```

```
SQL> CREATE TABLE GUIDES(Emp_ID NUMBER(3),  
2 Visitor_Phone NUMBER(10),  
3 CONSTRAINT FK_GUEMP  
4 FOREIGN KEY (Emp_ID) REFERENCES EMPLOYEE(Emp_ID) ON DELETE CASCADE,  
5 CONSTRAINT FK_GUDVST  
6 FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON DELE  
TE CASCADE,  
7 CONSTRAINT PK_GUEMPVST  
8 PRIMARY KEY (Emp_ID,Visitor_Phone));  
  
Table created.
```

6.12 < PARTICIPATES > Table

```
CREATE TABLE PARTICIPATES(Event_ID NUMBER(3),  
    Animal_ID NUMBER(3),  
CONSTRAINT FK_PRTEVN  
    FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE,  
CONSTRAINT FK_PRTANM  
    FOREIGN KEY (Animal_ID) REFERENCES ANIMAL(AnimalID) ON DELETE CASCADE,  
CONSTRAINT PK_PRTEVNANM  
    PRIMARY KEY(Event_ID,Animal_ID));
```

```
SQL> CREATE TABLE PARTICIPATES(Event_ID NUMBER(3),  
2     Animal_ID NUMBER(3),  
3     CONSTRAINT FK_PRTEVN  
4     FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE,  
5     CONSTRAINT FK_PRTANM  
6     FOREIGN KEY (Animal_ID) REFERENCES ANIMAL(AnimalID) ON DELETE CASCA  
DE,  
7     CONSTRAINT PK_PRTEVNANM  
8     PRIMARY KEY(Event_ID,Animal_ID));  
Table created.
```

6.13 < TICKET > Table

```
CREATE TABLE TICKET (TicketID NUMBER(3) PRIMARY KEY,  
    TicketName VARCHAR2(30),  
    TDate DATE,  
    Price NUMBER(5));
```

```
SQL> CREATE TABLE TICKET (TicketID NUMBER(3) PRIMARY KEY,  
2     TicketName VARCHAR2(30),  
3     TDate DATE,  
4     Price NUMBER(5));  
Table created.
```

6.14 < EVENT_TICKET > Table

```
CREATE TABLE EVENT_TICKET (EventTID NUMBER(3) PRIMARY KEY,  
                             Event_ID NUMBER(3),  
  
CONSTRAINT FK_EVNTCKT  
    FOREIGN KEY (EventTID) REFERENCES TICKET(TicketID) ON DELETE CASCADE,  
  
CONSTRAINT FK_EVNTCKE  
    FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE );
```

```
SQL> CREATE TABLE EVENT_TICKET (EventTID NUMBER(3) PRIMARY KEY,  
2                               Event_ID NUMBER(3),  
3   CONSTRAINT FK_EVNTCKT  
4       FOREIGN KEY (EventTID) REFERENCES TICKET(TicketID) ON DELETE CASCADE  
5  
6   CONSTRAINT FK_EVNTCKE  
7       FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE )  
;  
  
Table created.
```

6.15 < AREA_TICKET > Table

```
CREATE TABLE AREA_TICKET(AreaTID NUMBER(3) PRIMARY KEY,  
                           Area_ID NUMBER(3),  
  
CONSTRAINT FK_ARETCKA  
    FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE,  
  
CONSTRAINT FK_ARETCKT  
    FOREIGN KEY (AreaTID) REFERENCES TICKET(TicketID) ON DELETE CASCADE);
```

```
SQL> CREATE TABLE AREA_TICKET(AreaTID NUMBER(3) PRIMARY KEY,  
2                               Area_ID NUMBER(3),  
3   CONSTRAINT FK_ARETCKA  
4       FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE,  
5   CONSTRAINT FK_ARETCKT  
6       FOREIGN KEY (AreaTID) REFERENCES TICKET(TicketID) ON DELETE CAS  
CADE);  
  
Table created.
```

6.16 < HAS > Table

CREATE TABLE HAS (Bill_ID NUMBER(3),

Ticket_ID NUMBER(3),

Quantity NUMBER(3),

CONSTRAINT FK_HASBILL

FOREIGN KEY (Bill_ID) REFERENCES BILL(Bill_ID) ON DELETE CASCADE,

CONSTRAINT FK_HASTCK

FOREIGN KEY (Ticket_ID) REFERENCES TICKET (TicketID) ON DELETE CASCADE,

CONSTRAINT PK_HASBT

PRIMARY KEY(Bill_ID,Ticket_ID));

```
SQL> CREATE TABLE HAS (Bill_ID NUMBER(3),
2      Ticket_ID NUMBER(3),
3      Quantity NUMBER(3),
4  CONSTRAINT FK_HASBILL
5      FOREIGN KEY (Bill_ID) REFERENCES BILL(Bill_ID) ON DELETE CASCADE,
6  CONSTRAINT FK_HASTCK
7      FOREIGN KEY (Ticket_ID) REFERENCES TICKET (TicketID) ON DELETE
CASCADE,
8  CONSTRAINT PK_HASBT
9      PRIMARY KEY(Bill_ID,Ticket_ID));

Table created.
```

7 Constraints Script

Business Rule	SQL Script	Table
Each employee must work for one department, and each department must have at least one employee belonging to it.	<pre> CONSTRAINT FK_EMPDEP FOREIGN KEY (Dep_ID) REFERENCES DEPARTMENT(DepartmentID) ON DELETE CASCADE, </pre>	EMPLOYEE
An employee guides many visitors, and each visitor can be guided by one or more employees.	<pre> 1-CONSTRAINT FK_GUEMP FOREIGN KEY (Emp_ID) REFERENCES EMPLOYEE(Emp_ID) ON DELETE CASCADE 2-CONSTRAINT FK_GUDVST FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON DELETE CASCADE </pre>	Guides
An employee can work in one area, and each area must have many employees.	<pre> CONSTRAINT FK_EMPARE FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE </pre>	EMPLOYEE
Employee's age must be greater than or equal to 18 years.	<pre> CHECK (Age>18) </pre>	EMPLOYEE_AGE
An area must have several animals, and each animal must belong to one area.	<pre> CONSTRAINT FK_ANMARE FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE) </pre>	ANIMAL
Each event must have one location.	<pre> EventLocation VARCHAR2(50) NOT NULL </pre>	EVENT
A department controls several events, and each event must be controlled by a department.	<pre> CONSTRAINT FK_EVNDP FOREIGN KEY (DEP_ID) REFERENCES DEPARTMENT(DepartmentID) ON DELETE CASCADE); </pre>	EVENT
Each visitor must pay one or more bills, and each bill must be paid by one visitor .	<pre> CONSTRAINT FK_BILLVST FOREIGN KEY (Visitor_Phone) REFERENCES VISITOR(VisitorPhone) ON DELETE CASCADE, </pre>	BILL

An animal can participate in several events, and each event must have one or more animals participating in it.	1-CONSTRAINT FK_PRTEVN FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE, 2-CONSTRAINT FK_PRTANM FOREIGN KEY (Animal_ID) REFERENCES ANIMAL(AnimalID) ON DELETE CASCADE,	PARTICIPATES
Each bill must have a payment method, and each payment method can have one or more bills.	CONSTRAINT FK_BILLPM FOREIGN KEY (PMethod_ID) REFERENCES PAYMENT_METHOD(MethodID) ON DELETE CASCADE);	BILL
Each bill must have many tickets, and each ticket can have one or more bills.	1-CONSTRAINT FK_HASBILL FOREIGN KEY (Bill_ID) REFERENCES BILL(Bill_ID) ON DELETE CASCADE, 2-CONSTRAINT FK_HASTCK FOREIGN KEY (Ticket_ID) REFERENCES TICKET (TicketID) ON DELETE CASCADE,	HAS
An area must have several area tickets, and each area ticket can have only one area.	1-CONSTRAINT FK_ARETCKA FOREIGN KEY (Area_ID) REFERENCES AREA(Area_ID) ON DELETE CASCADE, 2-CONSTRAINT FK_ARETCKT FOREIGN KEY (AreaTID) REFERENCES TICKET(TicketID) ON DELETE CASCADE);	AREA_TICKET
An event must have several event tickets, and each event ticket can have only one event.	1-CONSTRAINT FK_EVNTCKT FOREIGN KEY (EventTID) REFERENCES TICKET(TicketID) ON DELETE CASCADE, 2-CONSTRAINT FK_EVNTCKE FOREIGN KEY (Event_ID) REFERENCES EVENT(EventID) ON DELETE CASCADE);	EVENT_TICKET

8 Queries

8.1 <Query 1>

Query in natural language (ENGLISH):

List all employee IDs and names who work in the same area as employee number 1.

SQL Script:

```
SELECT Emp_ID ,EmpFName || ' ' || EmpLName AS "Employee name"  
FROM EMPLOYEE  
WHERE Area_ID IN (SELECT Area_ID  
                  FROM EMPLOYEE  
                  WHERE Emp_ID =1);
```

Caption of the first five rows of the output:

EMP_ID	Employee name
1	Ahmed Saleh
15	Afyaa Faisal

8.2 <Query 2>

Query in natural language (ENGLISH):

Retrieve the number of visitors for each event.

SQL Script:

```
SELECT Ticket_ID AS "EVENT TICKET ID",SUM(Quantity) AS "Number of visitors"
FROM HAS , EVENT_TICKET ,EVENT
WHERE EVENT_TICKET.EventTID=HAS.Ticket_ID AND
EVENT_TICKET.Event_ID=EVENT.EventID
GROUP BY Ticket_ID
HAVING Ticket_ID IN (SELECT EventTID
                     FROM EVENT_TICKET );
```

Caption of the first five rows of the output:

EVENT TICKET ID	Number of visitors
1	1
6	6
11	5
13	3
8	2

8.3 <Query 3>

Query in natural language (ENGLISH):

List all employees' first names and ids who work in the same area that the lion belongs to with a birthdate of 25-MAY-2012.

SQL Script:

```
SELECT EmpFName ,Emp_ID
FROM EMPLOYEE
WHERE Area_ID IN (SELECT Area_ID
                  FROM ANIMAL
                  WHERE Type ='Lion' AND A_DateOfBirth='25-MAY-2012' );
```

Caption of the first five rows of the output:

EMPNAME	EMP_ID
Suha	14

8.4 <Query 4>

Query in natural language (ENGLISH):

Retrieve the animals ID and type who has been participated in any event that has been controlled by event management department

SQL Script:

```
SELECT AnimalID,Type
FROM ANIMAL , PARTICIPATES
WHERE ANIMAL.AnimalID =PARTICIPATES.Animal_ID AND
Event_ID IN (SELECT EventID
FROM EVENT ,DEPARTMENT
WHERE Dep_ID=DepartmentID AND DepartmentName='Event Management');
```

Caption of the first five rows of the output:

ANIMALID	TYPE
16	Crocodile

8.5 <Query 5>

Query in natural language (ENGLISH):

For each department that has more than 2 employees, retrieve the department ID and number of male employees .

SQL Script:

```
SELECT Dep_ID,count(*) AS "Number of male employees"
FROM  EMPLOYEE
WHERE Dep_ID IN(SELECT Dep_ID
                  FROM  EMPLOYEE
                  GROUP BY Dep_ID
                  HAVING count(*)>=2)
      AND E_Gender='Male'
GROUP BY Dep_ID;
```

Caption of the first five rows of the output:

DEP_ID	Number of male employees
116	1
118	2

9 Appendix

DEPARTMENTID	DEPARTMENTNAME	DEPARTMENTLOCATION
110	Animal Care	Building 1-first floor
111	Conservation and Research	Building 1-Second floor
112	Education	Building 1-third floor
113	Facilities and Operations	Building 2-third floor
114	Marketing and Development	Building 2-first floor
115	Veterinary Services	Building 2-Second floor
116	Finance and Administration	Building 3-Second floor
117	Curatorial	Building 3-first floor
118	Guidance	Building 3-first floor
119	Event Management	Building 3-third floor

Table 1 DEPARTMENT

EVENTID	EVENTNAME	EVENTLOCATION	DEP_ID
210	Dolphins show	Aquarium area	110
211	Zoo camp	Aquarium area	112
212	Night safaris	Predators area	110
213	Pets show	Pets area	110
214	Horse Race	Horse stable	110
215	Spring blossom festival	Butterflies garden	119
216	Animal feeding	Birds area	110
217	Animal feeding	Farm animals	110
218	crocodilians show	Reptile House	119
219	Pandas show	Pandas exhibit	110

Table 2 EVENT

AREA_ID	AREANAME	AREALOCATION
310	Aquarium	South area-section 1
311	Birds	South area-section 2
312	Reptile House	East area-section 1
313	Predators	East area-section 2
314	Farm animals	North area-section 1
315	Pets	North-section 2
316	Bats cave	Nest area-section 1
317	Butterflies garden	West area-section 2
318	Pandas exhibit	West area-section 3
319	Horse stable	North area-section 3

Table 3 AREA

ANIMALID	TYPE	A_GENDER	A_DATEOFB	AREA_ID
10	Dolphin	Female	10-NOV-21	310
11	Snake	Male	11-JUN-20	312
12	Chicken	Male	10-FEB-23	314
13	Bat	Male	20-APR-21	316
14	Panda	Female	16-NOV-19	318
15	Horse	Male	14-MAR-15	319
16	Crocodile	Male	10-MAY-12	312
17	Shark	Male	10-OCT-11	310
18	Flamingo	Female	23-NOV-22	311
19	Cat	Male	10-DEC-18	315
20	Dog	Female	29-APR-17	315
ANIMALID	TYPE	A_GENDER	A_DATEOFB	AREA_ID
21	Fox	Male	10-FEB-20	313
22	Lion	Male	25-MAY-12	313

Table 4 ANIMAL

E_DATEOFB	AGE
11-OCT-99	25
12-NOV-92	32
09-OCT-89	35
22-OCT-89	35
01-SEP-00	24
01-AUG-00	24
20-AUG-86	38
11-APR-99	25
12-APR-79	46
02-DEC-02	22

Table 5 EMPLOYEE_AGE

POSITION	SALARY
Zookeeper	2000
School Programs Coordinator	3000
Operations Supervisor	3500
Marketing Manager	6000
Zoo Veterinarian	5000
Finance Manager	8000
Accountant	9000
Tour Guide	2500
Safety Coordinator	10000
Social Media Coordinator	7000
Data Analyst	9000
POSITION	SALARY
Event Manager	8000

Table 6 EMPLOYEE_SALARY

EMP_ID	EMPFNAME	EMPLNAME	E_DATEOFB	POSITION	E_GENDER	E_PHONE	DEP_ID	AREA_ID
1	Ahmed	Saleh	11-OCT-99	Event Manager	Male	505781033	119	312
2	Omar	Saad	12-NOV-92	Marketing Manager	Male	555781130	114	
3	Noor	Ali	09-OCT-89	Tour Guide	Female	565784430	118	316
4	Khaled	Saad	22-OCT-89	Zookeeper	Male	556681132	110	318
5	Sara	Fahad	01-SEP-00	Accountant	Female	566891427	116	
6	Ghala	Mohammed	01-AUG-00	Zoo Veterinarian	Female	546981732	115	311
7	Khaled	Salem	20-AUG-86	Finance Manager	Male	566693132	116	
8	Fahad	Ahmad	11-APR-99	Safety Coordinator	Male	546811322	113	314
9	Lara	Naif	12-APR-79	Data Analyst	Female	567641932	111	
10	Saleh	Salem	02-DEC-02	School Programs Coordinator	Male	566387139	112	310
11	Ali	Talal	11-APR-99	Tour Guide	Male	565784930	118	311
EMP_ID	EMPFNAME	EMPLNAME	E_DATEOFB	POSITION	E_GENDER	E_PHONE	DEP_ID	AREA_ID
12	Sara	Anas	02-DEC-02	Tour Guide	Female	565994430	118	314
13	Lama	Ali	11-OCT-99	Tour Guide	Female	565994440	118	310
14	Suha	Anas	11-APR-99	Tour Guide	Female	565994477	118	313
15	Afyaa	Faisal	02-DEC-02	Tour Guide	Female	565943430	118	312
16	Lama	Abdulaziz	12-APR-79	Tour Guide	Female	545394430	118	315
17	Meaad	Ali	20-AUG-86	Tour Guide	Female	565994555	118	317
18	Wasayef	Abdullah	12-NOV-92	Tour Guide	Female	565994987	118	318
19	Khalid	Zafer	02-DEC-02	Tour Guide	Male	505763453	118	319

Table 7 EMPLOYEE

VISITORPHONE	VISITORFNA	VISITORLNA
505672987	Areej	Algrni
505612387	Sama	Alshamrani
505612388	Rashid	Alshahrani
505612389	Rahaf	Alshahrani
505612390	Gadeer	Alharbi
505612391	Sultan	Algrni
505612392	Mohanad	Alotaibi
505612393	Naif	Alotaibi
505612394	Salem	Alzahrani
505612395	Waled	Alkhthami

Table 8 VISITOR

METHODID METHODNAME

```

1 Cash
2 Mada Card
3 Visa Card
4 Tamara
5 Tabby

```

Table 9 PAYMENT_METHOD

BILL_ID	STATUS	TOTALPRICE	VISITOR_PHONE	PMETHOD_ID
1	Paid	50	505672987	1
2	Paid	30	505612387	2
3	Paid	40	505612388	2
4	Paid	25	505612389	2
5	Paid	30	505612390	3
6	Paid	90	505612391	2
7	Paid	30	505612392	5
8	Paid	40	505612393	4
9	Paid	45	505612394	1
10	Paid	90	505612395	3
11	Paid	250	505612395	1
BILL_ID	STATUS	TOTALPRICE	VISITOR_PHONE	PMETHOD_ID
12	Paid	300	505612394	2
13	Paid	30	505612389	2
14	Paid	360	505612389	2
15	Paid	105	505612394	3
16	Paid	210	505612391	2
17	Paid	125	505612392	5
18	Paid	80	505612389	4
19	Paid	70	505612387	1
20	Paid	195	505612390	3

Table 10 BILL

EMP_ID	VISITOR_PHONE
-----	-----
11	505612389
12	505612391
13	505612387
13	505612395
13	505672987
14	505612394
15	505612391
15	505612392
16	505612387
16	505612388
16	505612389
EMP_ID	VISITOR_PHONE
-----	-----
17	505612389
17	505612392
18	505612389
18	505612390
18	505612393
19	505612390
19	505612394

Table 11 GUIDES

EVENT_ID	ANIMAL_ID
-----	-----
210	10
211	10
211	11
213	19
213	20
214	15
216	18
217	12
218	16
219	14

Table 12 PARTICIPATES

TICKETID	TICKETNAME	TDATE	PRICE
1	Dolphins show	19-OCT-23	50
2	Aquarium	20-OCT-23	15
3	Pets show	25-OCT-23	10
4	Butterflies garden	20-NOV-23	25
5	Horse stable	12-DEC-23	15
6	crocodilians	07-JAN-23	15
7	Reptile House	20-SEP-23	10
8	Pandas show	20-AUG-23	20
9	Bats cave	03-AUG-23	15
10	Zoo camp	10-OCT-23	30
11	Dolphins show	06-FEB-23	50
TICKETID	TICKETNAME	TDATE	PRICE
12	Night safaris	10-APR-23	50
13	Pets show	25-APR-23	10
14	Birds	08-SEP-23	45
15	Horse stable	11-OCT-23	15
16	Farm animals	04-SEP-23	35
17	Spring blossom festival	14-MAR-23	25
18	Pandas show	17-AUG-23	20
19	Pets	03-MAR-23	35
20	Pandas exhibit	10-JAN-23	65

Table 13 TICKET

EVENTTID	EVENT_ID
1	210
3	213
6	218
8	219
10	211
11	210
12	212
13	213
17	215
18	219

Table 14 EVENT_TICKET

AREATID	AREA_ID
2	310
4	317
5	319
7	312
9	316
14	311
15	319
16	314
19	315
20	318

Table 15 AREA_TICKET

BILL_ID	TICKET_ID	QUANTITY
1	1	1
2	2	2
3	3	4
4	4	1
5	5	2
6	6	6
7	7	3
8	8	2
9	9	3
10	10	3
11	11	5
BILL_ID	TICKET_ID	QUANTITY
12	12	6
13	13	3
14	14	8
15	15	7
16	16	6
17	17	5
18	18	4
19	19	2
20	20	3

Table 16 HAS